7 EXAM PRACTICE

1 Which of the following solids consists of atoms or molecules held together by London forces?

A I_2

B Cu

C NaCl

D SiO_2 [1]

(Total for Question 1 = 1 mark)

2 Which of the following statements describes a phenomenon that can be explained by intermolecular hydrogen bonding?

A The boiling points of the alkanes increase with increasing relative molecular mass.

B CH_3COCH_3 (M_r = 58) has a higher boiling temperature than $CH_3CH_2CH_3$ (M_r = 44).

C Hydrogen chloride forms an acidic solution when dissolved in water.

D Ice has a lower density than water at 0 °C. [1]

(Total for Question 2 = 1 mark)

3 Which of the following isomers is likely to have the highest boiling temperature?

A $(CH_3)_2CHCH_2CH_2CH_3$

B $CH_3CH_2CH_2CH_2CH_2CH_3$

C $CH_3CH_2CH(CH_3)CH_2CH_3$

D $(CH_3)_2CHCH(CH_3)_2$ [1]

(Total for Question 3 = 1 mark)

4 Which quantity would best indicate the relative strengths of the hydrogen bond between the molecules in liquid hydrogen halides, HF, HCl, HBr and HI?

A bond enthalpies

B enthalpy changes of formation

C enthalpy changes of solution

D enthalpy changes of vaporisation [1]

(Total for Question 4 = 1 mark)

5 The surface of Triton, a moon of the planet Neptune, contains condensed methane that flows rapidly.

Which statement explains why condensed methane flows rapidly?

A Condensed methane has a metallic structure.

B Methane molecules contain strong covalent bonds.

C The intermolecular forces between methane molecules are weak.

D Methane molecules have a tetrahedral shape. [1]

(Total for Question 5 = 1 mark)

6 Which type of interaction is responsible for the intermolecular forces in liquid tetrachloromethane, CCl_4?

A covalent bonding

B hydrogen bonding

C instantaneous dipole-induced dipole attractions

D permanent dipole-permanent dipole attractions [1]

(Total for Question 6 = 1 mark)

7 Hydrogen bonding occurs in water and its presence has an effect on the properties of water.

(a) State what is meant by a hydrogen bond. [2]

(b) Draw a diagram to show the formation of a hydrogen bond between two water molecules. [2]

(c) State two ways in which the properties of water are affected by hydrogen bonding. [2]

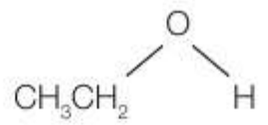
(Total for Question 7 = 6 marks)

8 Ethanol (CH_3CH_2OH) and methoxymethane (CH_3OCH_3) are structural isomers.

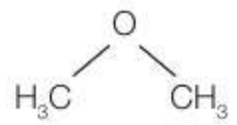
Ethanol is soluble in water whereas methoxymethane is not.

Their boiling temperatures are: ethanol, 78.5 °C; methoxymethane, –24.8 °C.

The structural formula and shape of each molecule is:



ethanol



methoxymethane

(a) Name the types of intermolecular interactions (forces) that exist in

(i) ethanol

(ii) methoxymethane. [5]

(b) Use your answers in part (a) to explain the difference in solubility and boiling temperatures of the two substances. [4]

(Total for Question 8 = 9 marks)

9 The table shows the relative molecular masses (M_r) and boiling temperatures of water and the first eight unbranched members of the homologous series of alkanes, methane to octane.

Formula of compound	H_2O	CH_4	C_2H_6	C_3H_8	C_4H_{10}	C_5H_{12}	C_6H_{14}	C_7H_{16}	C_8H_{18}
M_r	18	16	30	44	58	72	86	100	114
Boiling temperature / °C	100	-164	-89	-42	-0.5	36	69	98	125

(a) Comment on the difference in boiling temperature between water and methane. [3]

(b) Explain the trend in boiling temperatures of the alkanes. [3]

(c) Butane has an isomer called methylpropane, $(CH_3)_3CH$. Explain whether the boiling temperature of methylpropane is the same as, lower than or higher than that of butane. [3]

(Total for Question 9 = 9 marks)